

Sahar Soltanieh

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Professional Summary

Machine Learning Engineer with **5+ years** of experience designing and deploying self-supervised, multimodal, and federated deep learning systems. Strong track record spanning computer vision, time-series, and structured data, with published research and hands-on experience building production ML pipelines from messy, real-world data. Focused on translating research advances into robust, scalable engineering solutions with measurable impact.

Experience

The Hospital for Sick Children, Toronto, ON – Machine Learning Engineer Dec 2024 – Present

- **Built multimodal ML models** fusing 3D OCT, tabular OCT features, and clinical data for pediatric multiple sclerosis classification, improving diagnostic AUC by **7%** over best unimodal baseline.
- **Built automated ETL pipelines** linking MRI, OCT, VEP, demographics, and exercise data (**~2,000 patients**) into a centralized SQLite database, eliminating manual data merging and enabling reliable downstream analysis.

Current work (in progress):

- **Leading a federated learning initiative** across SickKids, Johns Hopkins, and Alberta Health Services, funded by a DSI Catalyst grant, enabling multi-institutional model training without centralizing sensitive patient data.
- **Engineered a layout-aware document extraction** pipeline for heterogeneous scanned patient records (notes, handwritten sections, tables, questionnaires), **routing to schema-specific models** – OCR, table-transformers, and Qwen2.5-VL; validated against neurologist-labeled ground truth via LLM-as-judge and human spot-checks.

Vector Institute, Toronto, ON – Machine Learning Associate Dec 2023 – May 2024

- **Delivered a demand-forecasting ML model** for ThisFish (seafood processors), translating industry stakeholder requirements into a deployed forecasting pipeline that reduced forecast error by **28%**.

AIIM Lab, Queen's University, Kingston, ON – Research Assistant Sep 2020 – May 2023

- **Designed a novel ECG distribution-analysis framework**, improving cross-dataset generalization by **16.8%**.
- **Implemented self-supervised representation learning** in PyTorch for ECG signals, improving downstream classification performance by **~10%** while using **90% less labeled data**.
- **Optimized contrastive-learning augmentations** for ECG self-supervised learning across 40 systematic augmentation-parameter configurations, improving downstream F1 score by **~20%** over baseline settings.

Projects

SEC Filings RAG Q&A Agent | (LangGraph, LangSmith, Chroma, SEC EDGAR MCP, Streamlit)

- Built a **RAG Q&A system** over **SEC 10-Q and 10-K** filings using **LangGraph** for orchestration, a **Chroma** vector database for retrieval, and **LangSmith** for observability, with a **Streamlit** front end for user interaction.
- Implemented **self-updating retrieval**: on a Chroma cache miss, queries the **SEC EDGAR MCP server** to fetch the missing filing and backfills the database, so subsequent queries on that filing are served locally.
- Designed a **difficulty-aware model router** that routes each query to the appropriate model tier, holding average cost to **\$0.10 per query**, evaluated the system against **60 synthetic samples** spanning multiple difficulty levels.

Selected Publications

- **Federated Learning in Neurology: Bridging Data Privacy and Artificial Intelligence for Brain Health**, Seminars in Neurology, 2026.
- **Multimodal machine learning for distinguishing pediatric multiple sclerosis from non-inflammatory conditions using optical coherence tomography**, Frontiers in Neurology, 2026.
- **In-distribution and out-of-distribution self-supervised ecg representation learning for arrhythmia detection**, Accepted in IEEE Journal of Biomedical and Health Informatics 2024.

Education

Queen's University – MSc in Artificial Intelligence Sep 2020 – May 2023

University of Tehran – BSc in Electrical Engineering Sep 2015 – Aug 2020

Technical Skills

Languages & Tools: Python, C, C++, SQL, Bash, Git, GitLab, Docker, Linux

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers, MONAI

Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, SQLite, DBeaver

AI Applications: Multimodal AI, Federated Learning, Self-Supervised Learning, Computer Vision (OpenCV, ViT), RAG & Prompt Engineering, Agentic AI (NL-to-SQL, LLM-based data validation)